Crazy Tens

A digital educational card game that reimagines Crazy Eights through dozenal (base-12) play, scoring, and visual learning.

Team #25 - The Crazy Four
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What is this game?

Crazy Tens is a web-based, two-player card game inspired by the familiar flow of Crazy Eights. Players still match cards by rank or suit, draw when they cannot play, and try to empty their hand first. The key difference is that the game is intentionally redesigned to expose players to dozenal thinking while they play.

Instead of treating the number system as separate classroom content, the software embeds it into the rules, score display, and interaction flow. The wild card is changed from 8 to 10 so that the symbol 10 directly signals a change in numeral-base meaning. Scores are also presented in dozenal notation, turning the end of each round into a lightweight number-base interpretation activity rather than a purely cosmetic display.

Why make this project?

- Decimal is historically dominant, but base-12 supports more convenient fraction structure because 12 is divisible by 2, 3, 4, and 6.
- Dozenal appears naturally in everyday life: time, dozens, months, inches, and many grouped quantities.
- Most learners never get an intuitive, playful way to compare decimal and dozenal in practice.
- A card game provides repeated decisions, visible scoring, and immediate feedback - ideal conditions for learning by doing.
- The project also serves as a complete software-engineering capstone, demonstrating architecture, real-time interaction, persistence, testing, and UI design.

Project goals

- Deliver a working digital game that remains recognizable to Crazy Eights players while adding base-12 meaning.
- Teach symbols, counting, divisibility, and score interpretation without demanding formal prior knowledge.
- Keep the interface accessible for casual users, but deep enough for students, educators, and technical reviewers.
- Build the system in a modular way so game rules, score logic, and number-base conversion can be tested and evolved independently.
- Produce a polished capstone artifact suitable for course review and EXPO demonstration.

How does gameplay work?

- A new session is created and cards are dealt to both players. The discard pile begins with a starter card.
- On each turn, a player chooses a legal card that matches either the active suit or the active rank.
- If no legal move exists, the player draws from the stock pile.
- Playing a 10 acts as a wild card and allows the player to declare a new suit through the interface.
- The system validates every move, rejects invalid plays, and gives immediate feedback.
- A round ends when one player empties their hand. The remaining cards in the opponent hand are summed and converted into dozenal for score display.
- When the stock pile is exhausted, the discard pile can be reshuffled to continue play.

Dozenal primer inside the game

Dozenal is base-12 rather than base-10. That means each place value represents powers of 12, and the symbols for ten and eleven are commonly written as A and B in software-facing explanations, or with dedicated dozenal glyphs in specialist contexts.

In Crazy Tens, the educational point is not to overwhelm the player with theory. Instead, the software exposes players to a few repeated ideas:

- 10 means "a full group of twelve" in base-12, not ten decimal units.
- Scores can be shown in both decimal and dozenal so players can compare representations.
- Counting and score reading become part of the play loop.
- Optional hints and tutorials can help novice users connect the symbol system to gameplay outcomes.

This makes the number system visible, practical, and memorable instead of abstract.

Who is it for?

- Students who have little or no prior exposure to dozenal and need a low-pressure introduction.
- Educators who want a compact digital activity that opens discussion about number bases, divisibility, and representation.
- Hobbyist card-game players who enjoy familiar mechanics with a mathematical twist.
- Technical reviewers and capstone assessors evaluating architecture, correctness, modularity, and user experience.
- Future contributors who may want to extend the project into new bases, new rules, or larger multiplayer experiences.

Educational value

- The game turns notation into action: players do not just read about base-12; they repeatedly encounter it in turns, symbols, and scoring.
- Because the play loop is familiar, learners can focus on the new representation instead of learning a completely new genre of game.
- The score display creates a natural reason to compare decimal and dozenal forms, making place value and conversion more concrete.
- Stretch features such as hints, onboarding primers, and arithmetic prompts can increase learning depth without making the MVP too complex.
- This aligns the project with a broader educational aim: showing that alternate numeral systems can be practical, intuitive, and worthy of exploration.

System design and modules

- Frontend modules include the application shell, authentication client, lobby view, game board, move controller, scoreboard view, and profile view.
- Backend modules include the HTTP API, real-time gateway, matchmaking, authentication, repository, and audit logging.
- Core game logic is separated into a game engine, rules module, scoring module, base-conversion module, and game-actions module.
- This separation matters because rules, scoring, and numeral-display policy may change after playtesting and should not require rewriting the whole system.
- The architecture supports a clean boundary between presentation, transport, persistence, and deterministic game logic.

Implementation approach and future directions

- Planned stack: React with TypeScript on the frontend, Node.js with TypeScript on the backend, a relational database such as PostgreSQL, and WebSocket-based real-time updates.
- Engineering workflow emphasizes GitHub issue tracking, Kanban planning, code review, CI/CD, and testing of both logic and interaction.
- The MVP focuses on two-player gameplay, core rules, score calculation, and dozenal display.
- Future work can include 3-4 player support, richer onboarding, optional hints, alternate numeric bases such as octal or hexadecimal, and broader educational analytics.
- The long-term vision is not only a single game, but a reusable model for teaching number systems through interactive play.

Conclusion: Crazy Tens combines a familiar card-game structure with a purposeful numerical twist. By linking rule changes, score reading, and interface feedback to dozenal representation, it aims to make base-12 understandable through play rather than through lecture alone.

Crazy 1-0's

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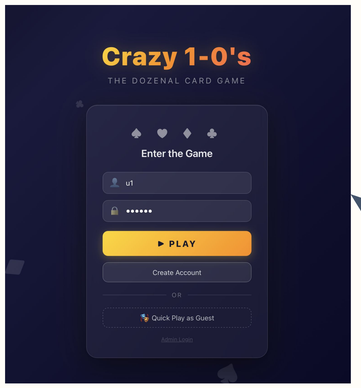
Dozenal primer inside the game

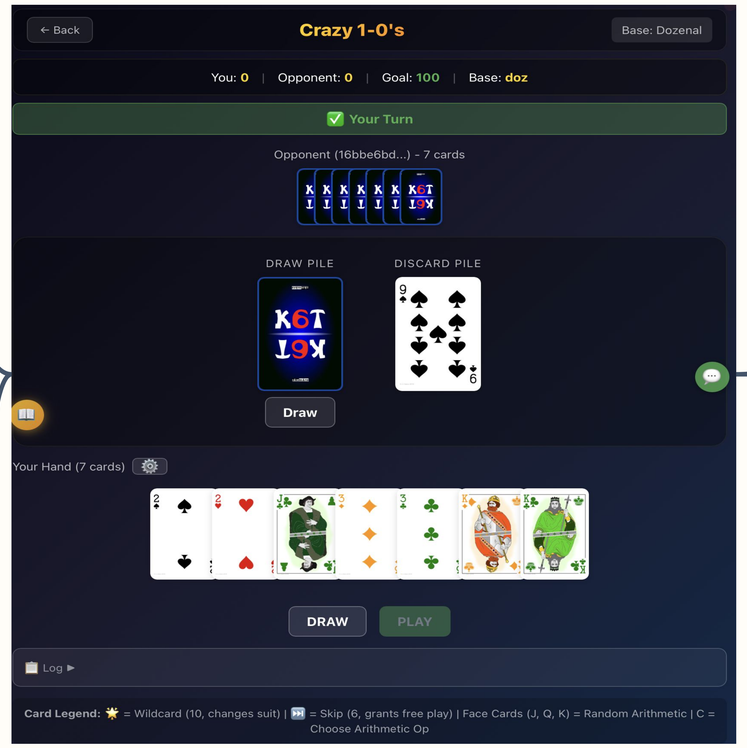
Dozenal is base-12, meaning place values are powers of 12. Symbols A and B are often used for ten and eleven. Crazy Tens avoids complex theory, instead exposing players to core ideas: 10 represents "a full group of twelve," scores are shown in both decimal and dozenal for comparison, counting is integrated into the play loop, and optional hints connect symbols to gameplay. This makes dozenal practical and memorable, not abstract.

How does gameplay work?

- Match cards by suit or rank; draw if no legal move exists.
- "10" is a wild card used to declare new suits.
- The system validates moves and provides instant feedback.
- Rounds end when a hand is empty; opponent cards are summed in dozenal.
- Exhausted stock piles are reshuffled to maintain continuous play.

USER LOGIN





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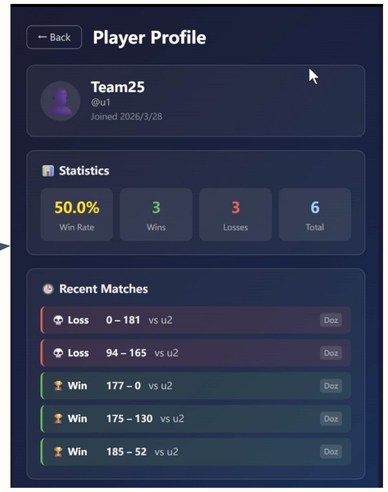
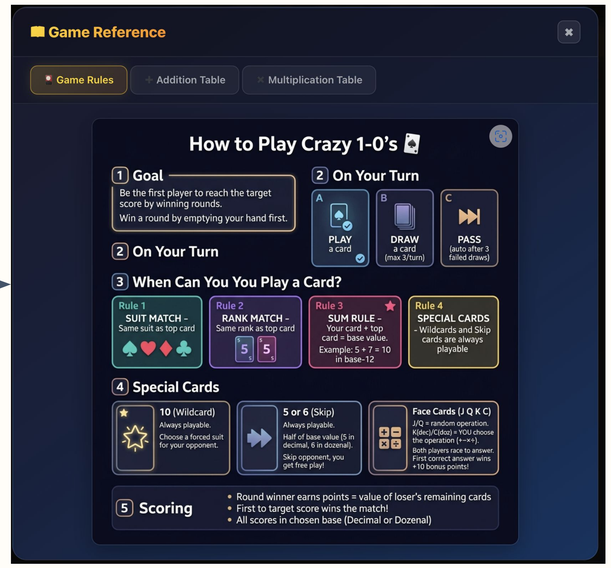


IN GAME RULE TOOLTIP



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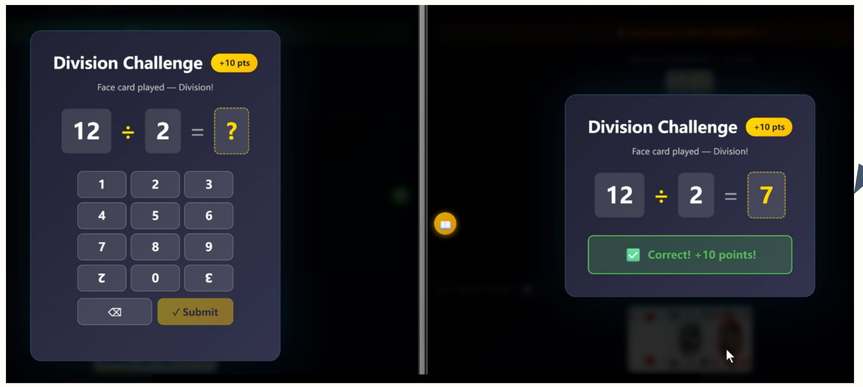
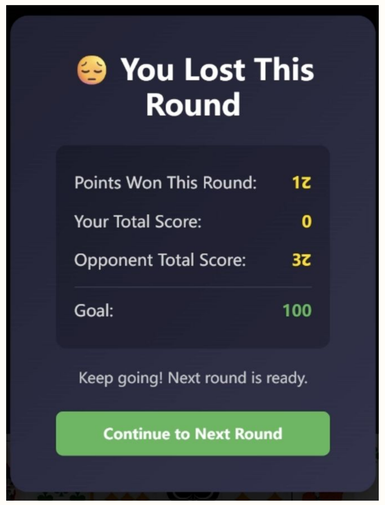


ROUND RECAP



Educational value

- Turns notation into action through repetitive gameplay loops.
- Familiar Crazy Eights mechanics allow focus on new representations.
- Score displays encourage concrete comparison of decimal and dozenal.
- Hints and primers increase depth without added complexity.
- Shows that alternate numeral systems are practical and intuitive.

System design and modules

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- The architecture supports a clean boundary between presentation, transport, persistence, and deterministic game logic.

Implementation approach and future directions

- Stack: React/TypeScript, Node.js, PostgreSQL, and WebSockets.
- Workflow: Kanban, CI/CD, and logic-focused unit testing.
- Future: 4-player support, octal/hex bases, and educational analytics.
- Vision: A reusable model for teaching systems through interactive play.

Crazy Tens makes base-12 understandable through play by linking rule changes and interface feedback directly to dozenal representation.